

Vaidehi Karve

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Education

University of California San Diego

San Diego, CA

Bachelor of Science in Data Science (Concentration: ML/AI), Minor in Business Analytics

Expected Graduation: June 2027

- **GPA: 3.86/4.0** (Provost Honors)
- **Coursework:** Data Structures & Algorithms (Python/Java), SQL & Database Management, Supervised/Unsupervised Machine Learning, ML Optimization, Business Analytics, Data Visualization, Scalable Analytics Systems, Distributed Computing

Work Experience

Data Science Alliance

San Diego, CA

Data Analyst Intern | \$2,250 UC San Diego Research Scholarship Winner 2025

Aug 2025 – Present

- Developed the Food Rescue Routing & Impact Tracker to optimize surplus food distribution at a San Diego food bank
- Built KPI dashboards in Tableau, sharing insights with stakeholders to guide food recovery decisions to serve 400K people monthly
- Designed vehicle routing models integrating donation and capacity data with Google Maps APIs for real-time pickup scheduling
- Measured ROI and operational impact (meals served, CO2 saved, pounds rescued) to support food recovery efforts and reduce cost

UC San Diego Health, FAIR Data Informatics (FDI) Lab

San Diego, CA

Data Engineering Intern

Oct 2024 – Present

- Built Python/FastAPI data pipelines to standardize and align 300+ biomedical variables across 5 datasets, improving interoperability
- Applied Hugging Face transformers (BioBERT) with REST APIs and Elasticsearch to automate semantic variable matching across biomedical datasets, reducing manual review by 20% and accelerating adoption of AI-ready biomedical datasets
- Produced FAIR-compliant datasets, strengthening data governance and enabling reproducible analytics across global research teams
- Evaluated LLM-based retrieval methods (ChatGPT, Hugging Face) and tested chatbot-style query assistants for dataset navigation

San Diego Gas & Electric

San Diego, CA

Data Science and Machine Learning Intern | Enterprise Data & Governance, Wildfire Analytics Team

June – Sept 2025

- Developed a forecasting models (XGBoost, regression) to predict hourly vehicle-pole collision risk, a wildfire ignition source
- Increased model accuracy by 5% by processing 14+ years of outage data (257K+ poles) in AWS Athena and integrating GIS features
- Built a Power BI map dashboard of 5K+ poles by wildfire risk, enabling stakeholders to prioritize infrastructure investments and support executive decision-making regarding high-risk poles

AI4ALL Ignite Accelerator [\[Github\]](#)

Remote

AI/ML Fellow

May – Aug 2025

- Selected for a competitive 13-week fellowship in applied machine learning and generative AI system development
- Explored LLM workflows and ethical AI adoption; led a 5-person team to present results to 30+ stakeholders on AI applications
- Trained SVM and regression models on 3,700 MRI scans with 98.5% accuracy and 98.2% recall after SMOTE and cross-validation

Projects

U.S. Power Outage Analysis [\[Github\]](#) | Python, SQL, Time Series Modeling, scikit-learn

- Built a predictive modeling pipeline analyzing 1,500+ outages over 15 years, improving duration forecasts by 25%
- Engineered features and performed fairness analysis to ensure robustness across geography, weather, and population density
- Published an interactive web report with findings to support utility providers and regulators in grid reliability planning

Music Analysis and Recommender System | Python, NLP, Power BI

- Analyzed 250+ songs using NLP and 20+ audio features to identify trends in popularity and style evolution
- Built interactive Power BI dashboards to visualize listening trends and compare recommendations using experiment-style analysis
- Developed a search and recommendation engine integrating data ingestion, analysis, and visualization personalized to each user

Leadership Experience

Data Science Student Society (DS3), UC San Diego

May 2024 – Present

Events Director (Executive)

- Managed \$10K+ budget and logistics for 50+ events annually, engaging 500+ active student members
- Collaborated with peers across technical, design, and events teams to plan and deliver large-scale workshops and hackathons

Skills

Programming & Data Tools: Python (pandas, NumPy, scikit-learn), SQL, R, Java, Git/GitHub

ML & AI: Supervised/Unsupervised Learning, Model Evaluation, Feature Engineering, NLP, Transformers, LLM, TensorFlow, PyTorch

Data Engineering: ETL/ELT Pipelines, AWS (Glue, S3, Athena), FastAPI, dbt, PySpark, Distributed Computing (Hadoop/MapReduce)

Analytics: Power BI, Tableau, Statistical Analysis, A/B Testing, KPI Dashboards, Data Storytelling, Time Series Forecasting

Systems & Collaboration: Scalable Analytics Systems, REST APIs, Agile/Scrum Methodologies, Experimentation Workflows